SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Case Study

Course Code:	ITA05	**Course Title:**	Computer Vision
CO Assessed:	**CO3 – Precision (accurate feature extraction & analysis) (BL3)**	**Assessment:**	**Assessment Tool 2 – Case Study**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)*		
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Section / Batch:	CSE	**Date:**	August 11, 2026

Code of Conduct

I, G. Tejasree *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: G. Tejasree

Criteria	Weight tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	/25
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	/25
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	/20
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	/20
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	/10
					Total Marks	/100

3. HOUGH TRANSFORM IN NOISY IMAGES

Case Study – Analysis and Improvement

Scenario: A system uses the Hough Transform to detect lines in noisy images but produces inaccurate results. The objective is to analyze the effect of noise on detection and suggest improvements based on preprocessing and parameter selection.

1. Effect of Noise on Hough Transform

The Hough Transform detects lines by mapping edge pixels from image space to parameter space. When an image contains noise, unwanted pixels can also be treated as edges. These pixels cast incorrect votes in the Hough accumulator, producing false peaks and inaccurate line detections.

- False line detections: Noise creates extra edge points that produce false lines.
- Broken or fragmented lines: Noise can interrupt real edges and make continuous lines appear discontinuous.
- Poor localization: Detected lines may be shifted away from their actual positions.
- Multiple accumulator peaks: Random noise produces many unnecessary votes.
- Increased computation: More edge points increase the number of votes and processing effort.

2. Key Problems in the Detection Process

Problem	Effect
High noise level	Creates many false edge points and false Hough votes.
Weak or broken edges	True lines receive fewer votes and may not be detected.
Low contrast	Makes the boundaries of important lines difficult to identify.
Improper threshold	Either too many false lines or missed real lines.
Poor resolution settings	Can reduce line accuracy or increase computation.

3. Recommended Preprocessing

1. Noise reduction: Apply a median filter for salt-and-pepper noise or a Gaussian filter for Gaussian/high-frequency noise.
2. Contrast enhancement: Use histogram equalization or CLAHE when the image has poor contrast.
3. Edge detection: Apply the Canny edge detector to obtain thin and well-localized edges.
4. Morphological processing: Use suitable opening/closing operations to remove small unwanted components or connect broken edges.

4. Parameter Selection and Tuning

Parameter	Purpose	Suggested Improvement
ρ (rho) resolution	Controls distance resolution in pixels.	Use a sufficiently small step, such as 1–2 pixels, for accurate detection.
θ (theta) resolution	Controls angular resolution.	Use a fine angular step when precise line orientation is required.
Threshold	Minimum accumulator votes needed to accept a line.	Increase it to suppress false lines, but avoid values that remove true lines.
Minimum line length	Rejects short detected segments.	Increase it to remove short noise-induced lines.
Maximum line gap	Controls how far apart segments may be connected.	Choose a value that reconnects real broken lines without joining unrelated segments.

5. Improved Detection Pipeline

Input noisy image

→ Noise reduction (Median/Gaussian filter)

→ Contrast enhancement (Histogram Equalization/CLAHE)

→ Canny edge detection

→ Hough Transform with optimized parameters

→ Remove false/duplicate lines

→ Final accurate line detection

6. Why These Improvements Work

Preprocessing reduces the number of unwanted edge pixels before the Hough Transform is applied. This makes the accumulator cleaner and allows genuine lines to receive stronger relative votes. Proper threshold, line-length, gap, and resolution settings then prevent weak noise-induced detections while retaining important lines.

7. Conclusion

Noise significantly affects Hough Transform performance by introducing false votes, fragmented edges, false lines, and inaccurate line locations. The problem can be reduced by applying appropriate denoising, contrast enhancement, and reliable edge detection before the transform. Careful selection of ρ and θ resolution, accumulator threshold, minimum line length, and maximum line gap further improves accuracy. Therefore, a combination of preprocessing and parameter tuning provides a more reliable Hough Transform-based line detection system.